Scalability Analysis & Future Engineering Roadmap

Date: 31 January 2026	**Team ID:**
Project: Global Electric Vehicle Market Performance Analysis	**Institution:** Medi-Caps University

1. Current System Limitations:

S.No	System Limitation	Operational Impact	Priority
1	Batch Ingestion via Excel files	New vehicle releases require manual batch reload rather than real-time stream.	High
2	Static Tableau Public embed links	Requires manual republishing if underlying Snowflake views undergo schema migration.	Medium
3	Absence of real-world ambient weather data	Range autonomy models do not yet simulate severe sub-zero winter degradation.	Medium

2. Scalability Architecture:

Aspect	Current State	Proposed Cloud Scalability Upgrade
Data Storage	200k rows in Snowflake Single-Cluster	Scale to billions of rows with Snowflake multi-cluster auto-scaling warehouses and Apache Iceberg open table formats.
Data Streaming	Batch Excel / CSV upload via COPY INTO	Snowpipe integration with Kafka / AWS Kinesis for real-time CAN-bus IoT vehicle telematics streaming.
BI & Serving	Tableau Public + Local Flask Server	Containerized Flask on Kubernetes (EKS/GKE) backed by Redis caching and Tableau Server REST API.
Predictive AI	Historical descriptive analytics	Deploy XGBoost / Scikit-learn models directly in Snowflake Snowpark to predict real-time battery degradation.